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CSS234 WEB PROGRAMMING II

LECTURE 09 WEB TESTING

Introduction

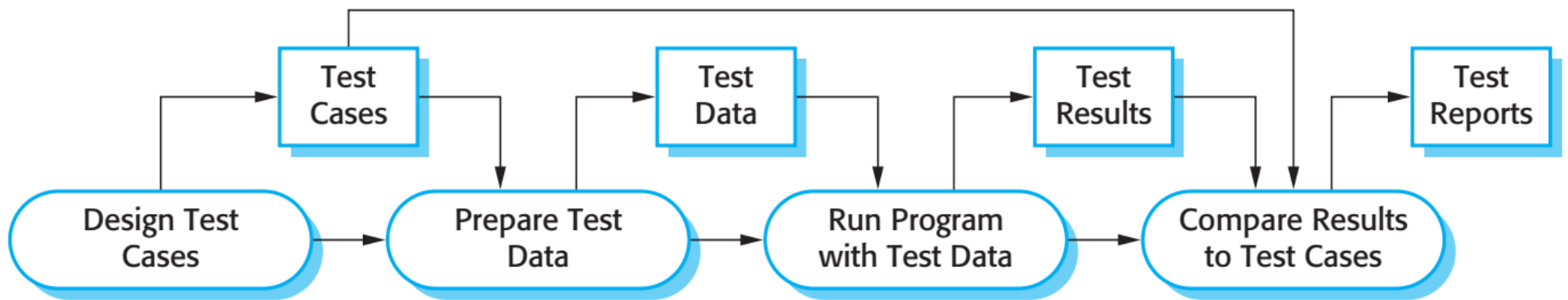
- Windows 2000 was said by some to contain a whopping 63,000+ known bugs when it was shipped
 - bugs per thousand lines of code (KLOC) is typically estimated at about 15 to 50
 - Windows 2000 contained more than 29 million lines of code
 - How many bugs in window 2000?
- Windows 8 - 80 million lines of code
- Linux operating system - 50 million lines of code
- Mac OS X Tiger has approximately 85 million lines of code
- Facebook has approximately 62 million lines of code

Testing Goals

- To see whether it **meets the requirements** and whether it **works correctly** under **all circumstances** (with any set of inputs)
- To remove all the bugs?.....
- To reduce **the number of bugs** and their **frequency of occurrence** so that users can get their jobs done with a minimum of annoyance

Testing process

- An abstract model of the traditional testing process



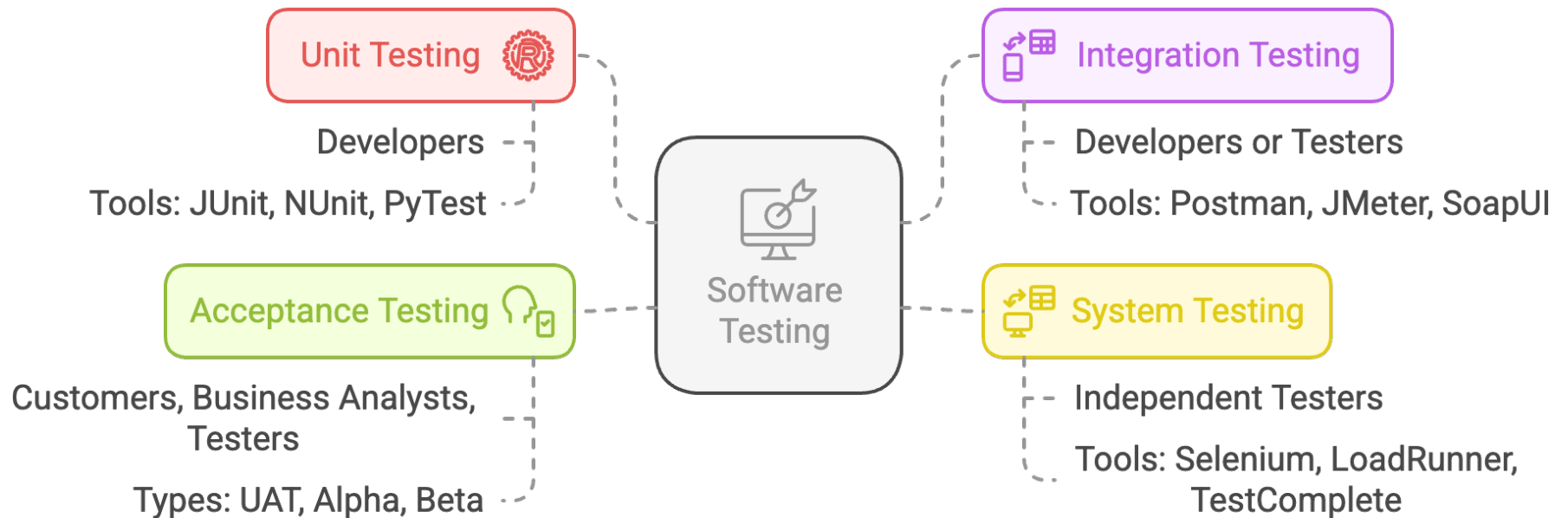
Bugs never die



Which bugs to fix

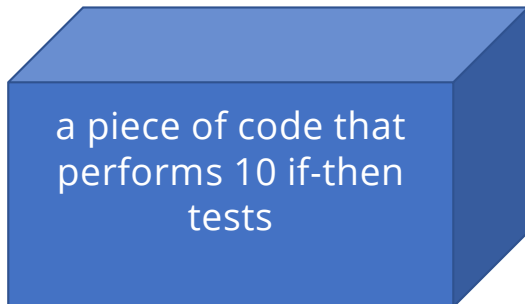
- **Severity**—How painful is the bug for the users?
How much work, time, money, or other resources are lost?
- **Workaround**—Are there workaround?
- **Frequency**—How often does the bug occur?
- **Difficulty**—How hard would it be to fix the bug?
- **Riskiness**—How risky would it be to fix the bug? If the bug is in particularly complex code, fixing it may introduce new bugs.

Testing Levels



Testing Levels - Unit testing

- A unit test verifies the correctness of a specific piece of code.
- As soon as you finish writing a piece of code, you should test it.
- Test it as thoroughly as possible because it will get harder to fix later.



- A piece of code that performs 10 if-then tests
- You need to test every possible combination of branches. For 10 branches with 2 paths each, that's $2^{10} = 1,024$ possibilities.
- Break the big piece of code into 10 pieces: making a total of 20 test cases

Testing Levels - Integration testing

- An integration test verifies that **the new method works and plays well with others.**
- It checks that existing code calls the new method correctly, and that the new method can call other methods correctly.
- Integration typically focuses on
 - the new code and other pieces of code that interact with it
 - verifying that the new code didn't mess up anything that seems unrelated

Testing Levels - Integration testing

- Ideally, when you finish unit testing a piece of code, you would then perform integration testing to make sure it fits in where it should and that it didn't break anything obvious.
- Then you perform regression testing to see if it broke something non-obvious.
- **Regression testing:** test the program's entire functionality to see if anything changed when you added new code to the project



Testing Levels - System testing

- System testing is an end-to-end run-through of the whole system.
- Exercises every part of the system to discover as many bugs as possible.
- For complicated applications, you'll probably have to test the most common and most important scenarios and leave some combinations untested.

Testing Levels - Acceptance testing

- The goal of acceptance testing is to determine whether the **finished application meets the customers' requirements.**
- Customer representative sits down with the application and runs through all the user cases you identified during the requirements.
- Requirements changed; mistake is the wait until the application finished before starting acceptance testing

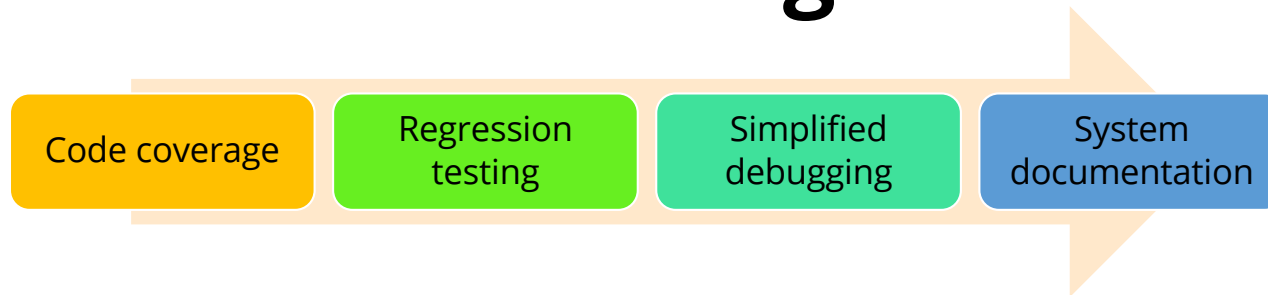
Automated testing

- Automated testing tools let you define tests and the results they should produce.
- After running a test, the testing tool can compare the results it got with expected results; some tools can even compare images to see if a result is correct.
- For example, to test a drawing program
 - you might record your actions as you draw, resize, and color a polygon.
 - Later the testing tool would repeat the steps you took

Load test

- **Load tests** that simulate a lot of users all running simultaneously to measure performance.
- Advantages:
 - Run overnight (no need to monitor)
- **OUTSOURCING TESTS**
 - software development team would write code during the day
 - perform their unit and integration tests, and then ship their code to testers in India before they left for the day

Testing



- Code coverage - every code segment that you write should have at least one associated test.
- Regression testing - to check that changes to the program have not introduced new bugs.
- Simplified debugging - When a test fails, it should be obvious where the problem lies. The newly written code needs to be checked and modified.
- System documentation - The tests themselves act as a form of documentation that describe what the code should be doing.

Release testing

- Release testing is the process of testing a particular release of a system that is intended for use outside of the development team (for customers and users)
 - A **separate team** that has not been involved in the system development should be responsible for release testing.
 - System testing by the development team should focus on discovering bugs in the system (defect testing).
 - The objective of release testing is to check that the system **meets its requirements** and **is good enough for external use** (validation testing)

Release testing

- **Requirements-based testing**

- A general principle of good requirements engineering practice is that requirements should be testable

- **Scenario testing**

- an approach to release testing where you devise typical scenarios of use and use these to develop test cases for the system. A scenario is a story that describes one way in which the system might be used.

MHC-PMS

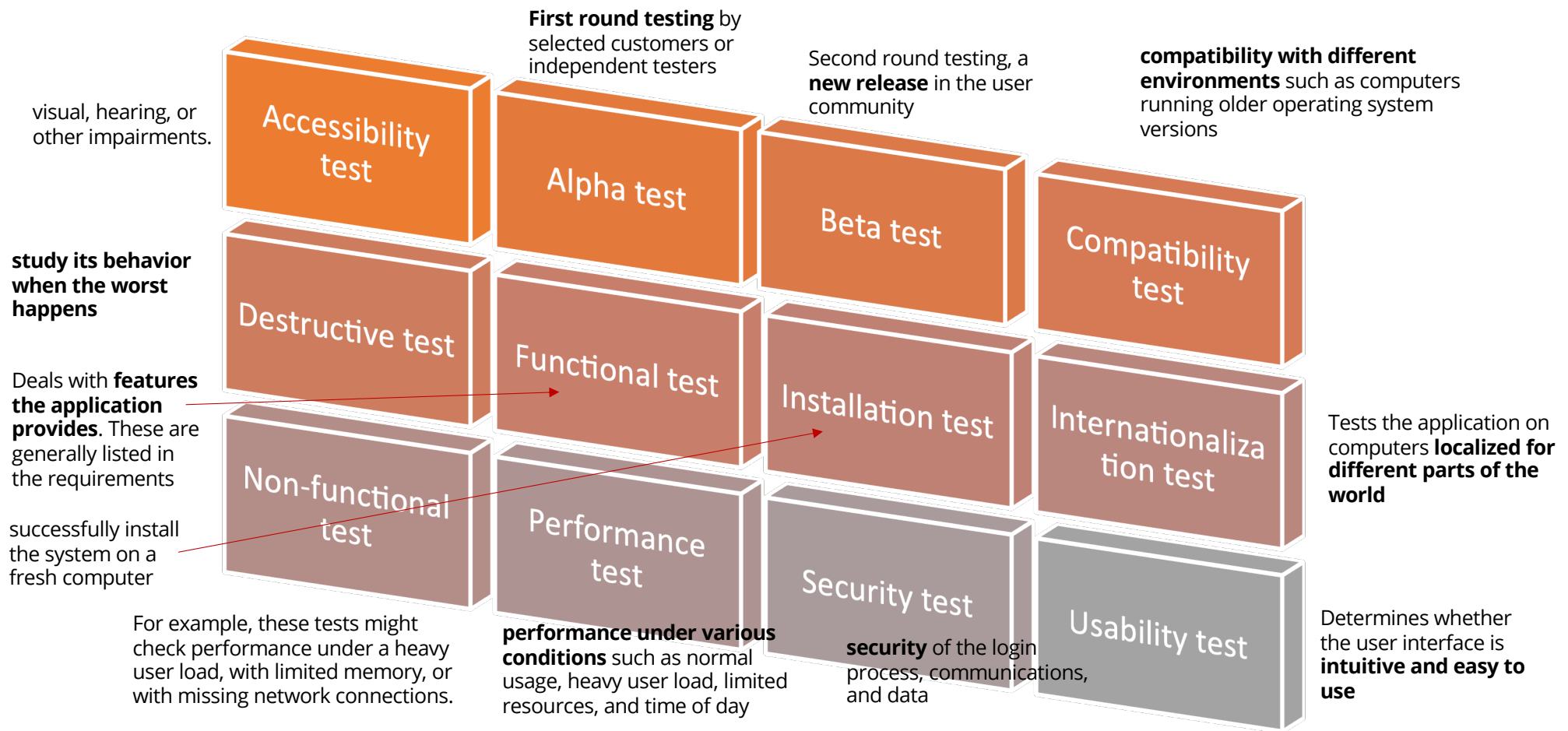
1. Authentication by logging on to the system.
2. Downloading and uploading of specified patient records to a laptop.
3. Home visit scheduling.
4. Encryption and decryption of patient records on a mobile device.
5. Record retrieval and modification.
6. Links with the drugs database that maintains side-effect information.
7. The system for call prompting.

Release testing

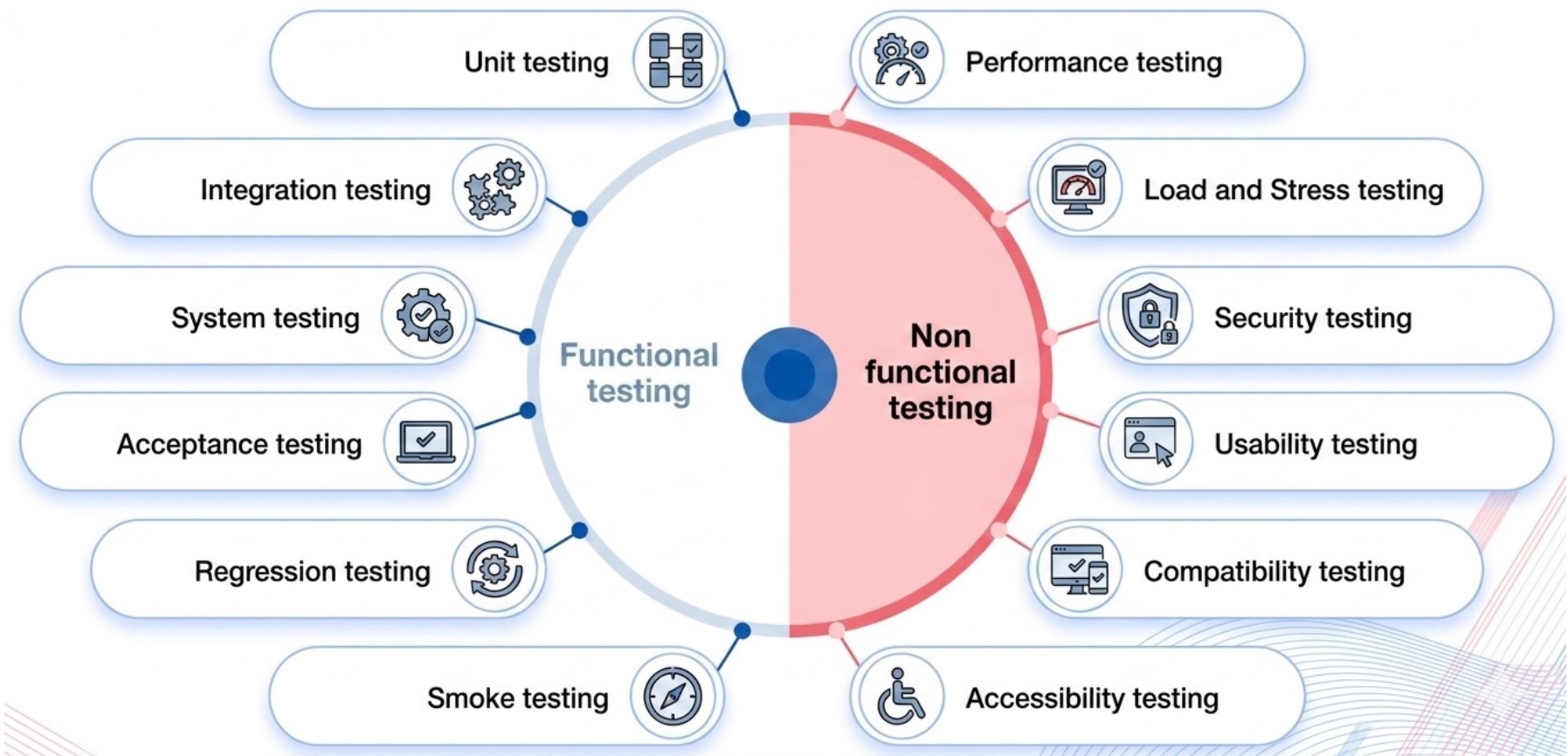
- Performance testing
 - It tests the failure behavior of the system e.g., system exceeds the maximum loads
 - It stresses the system and may cause defects to come to light that would not normally be discovered
 - Stress testing is particularly relevant to distributed systems based on a network of processors.
 - These systems often exhibit severe degradation when they are heavily loaded.
 - Stress testing helps you discover when the degradation begins so that you can add checks to the system to reject transactions beyond this point.

User testing – alpha, beta and acceptance testing

Other testing types



Types of web application testing





- Some people argue that developers should not be involved in testing their own code but that all testing should be the responsibility of a separate team.
- **Give arguments for and against testing by the developers themselves.**

Testing Techniques

- Exhaustive Testing – too many inputs
 - Tic-tac-toe : there are only $9! = 362,880$ possible board arrangements
- Black-Box Testing
- White-Box Testing
- Gray-Box Testing

Testing Techniques - Black-Box Testing

- You pretend the method is a black box that you can't peek inside.
- You know what it is supposed to do, but you have no idea how it works.
 - Sending it a bunch of random inputs (throw a few million random pairs of values at the Maximum method)
 - Even if you don't know how the method works, you can try to guess values that might mess it up
 - special values like 0 for numbers and blank for strings
 - the largest and smallest possible values
 - all blanks
 - special characters such as &#%!\$
 - look at boundary values, e.g., If a method expects a floating point parameter between 0.0 and 1.0, make sure it can handle those two values

Testing Techniques - White-Box Testing

- In white-box testing, you get to know how the method does its work.
- You then use your extra knowledge to design tests to try to make the method crash and burn
- **Disadvantage:** you know how the method works, so you might skip some test cases that you assume work.

Testing Techniques - White-Box Testing

- “zero-length strings” confusing the program, so you wrote a code to handle it.. So, you don’t test it.
 - Incorrectly handle.. And not testing it..
- Use white-box testing to create tests you know will be troublesome, but don’t skip tests that you “know” the method can handle.

Testing Techniques - Gray-Box Testing

- Gray-box testing is a combination of white-box and black-box testing
- Program: a method examines test score data to find students that might need extra tutoring help.
- You don't know all the details, but you do know that it uses the **quicksort** algorithm to sort the students by their grades
 - Because you don't know what else is going on inside the method, you also need to write a bunch of black-box styles tests.

How to fix a bug

- Ask yourself how you could prevent a similar bug in the future
- Ask yourself if a similar bug could be lurking somewhere else
- Look for bugs hidden behind this one
- Examine the code's method and look for other possibly unrelated bugs
- Make sure your fix doesn't introduce a new bug

Writing report!

1) Having a clearly specified Bug Number: Always assign a unique number to each bug report.

2) Reproducible: If your bug is not reproducible, then it will never get fixed. You should clearly mention the steps to reproduce the bug.

3) Be Specific

- Important Features In Your Bug Report
- 1) Bug Number/id
- 2) Bug Title
- 3) Priority When should a bug be fixed? Priority is generally set from P1 to P5. P1 as “fix the bug with the highest priority” and P5 as “Fix when time permits”.
- 4) Severity
- 5) Platform/Environment
- 6) Description
- 7) Steps to Reproduce
- 8) Expected and Actual Result
- 9) Screenshot

Severity: This describes the impact of the bug.

Types of Severity:

- **Blocker:** No further testing work can be done.
- **Critical:** Application crash, Loss of data.
- **Major:** Major loss of function.
- **Minor:** Minor loss of function.
- **Trivial:** Some UI enhancements.
- **Enhancement:** Request for a new feature or some enhancement in the existing one.

Example of the report

- Write detailed defect report for this sample defect:
After logging into Gmail, it navigates to Google.com

Defect: After logging into Gmail, it navigates to Google.com

Title – Gmail login navigates to Google home page rather than mail inbox.

Severity : High

Priority : Critical

Observed on : Windows – FF x.x and IE x.x and Mac – Chrome x

Module : Login

Reproduction Steps :

1. Launch a Gmail link with compatible browsers.
2. Click on Sign in button in link, Gmail login window screen is appeared.
3. Enter a valid login credentials.
4. Click on SIGN IN button,

Analysis :

Defect Status : Open

Assigned to : Project Manager x.x